

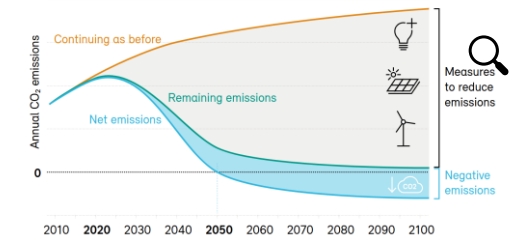
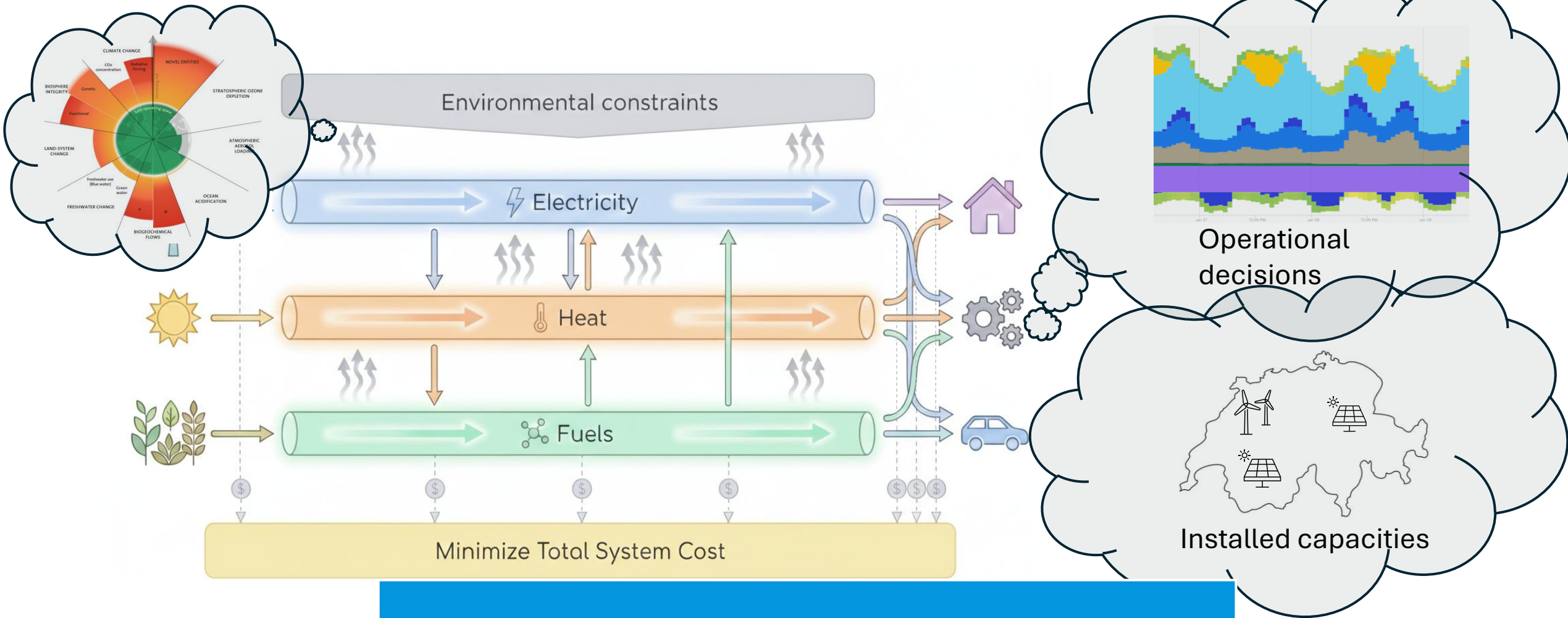
Exploration of Near-Optimal Pathways for the Energy Transition

Evren M. Turan, Stefano Moret, André Bardow



Sector-coupled energy system models

What are they?



Is the cost-optimal system what we want?

No: what are the options and trade-offs



1. All models are wrong...

- *The optimal solution is not the real-world optimal*

2. The objective is misaligned...

- *Socially not preferred*
- *Unmodelled burden shifts*

Trutnevyte, E. (2016). Does cost optimization approximate the real-world energy transition?. *Energy*, 106, 182-193

Tröndle, T., Mey, F., & Lilliestam, J. (2025). Socially preferable and technically feasible: European citizens choose solar power and import independence over lower costs. *Energy Research & Social Science*, 129, 104364.

de Tomás-Pascual, Alexander, et al. (2025). The optimal is not always the best: Life cycle impacts of near-optimal energy systems. *Applied Energy*, 399, 126487.

Stigler, G. J. (1945). The cost of subsistence. *Journal of farm economics*.

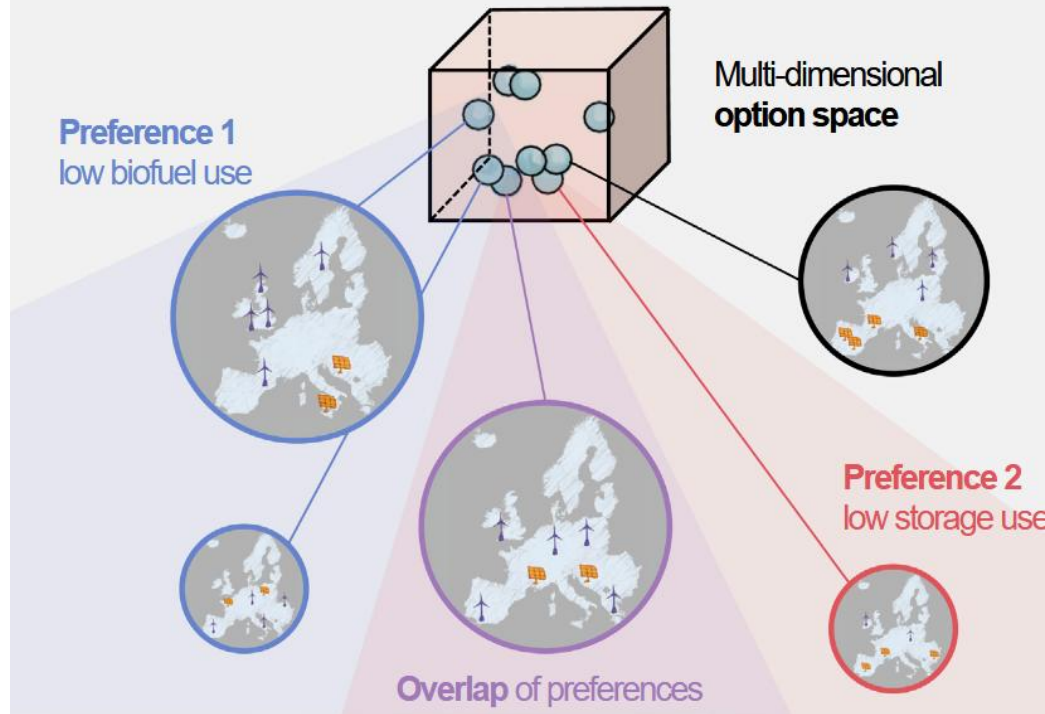
Near-optimal solutions

Define a cost budget, and see what are the options

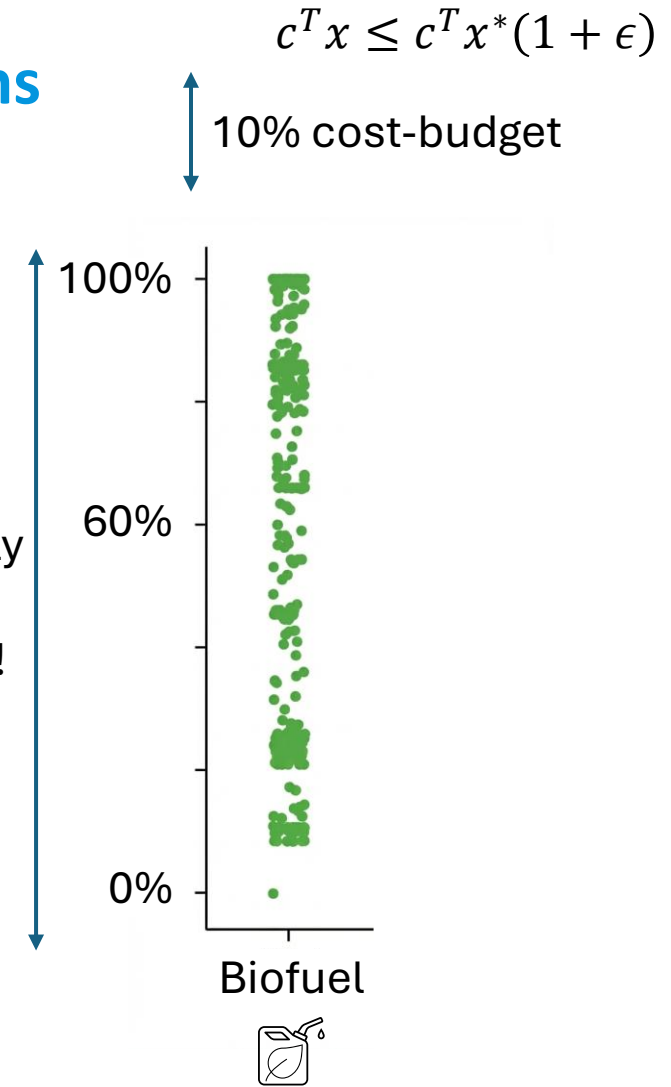
Why 441?

441 technically feasible and cost-effective options (●)
for an energy self-sufficient, carbon-neutral Europe

Why do options = points?



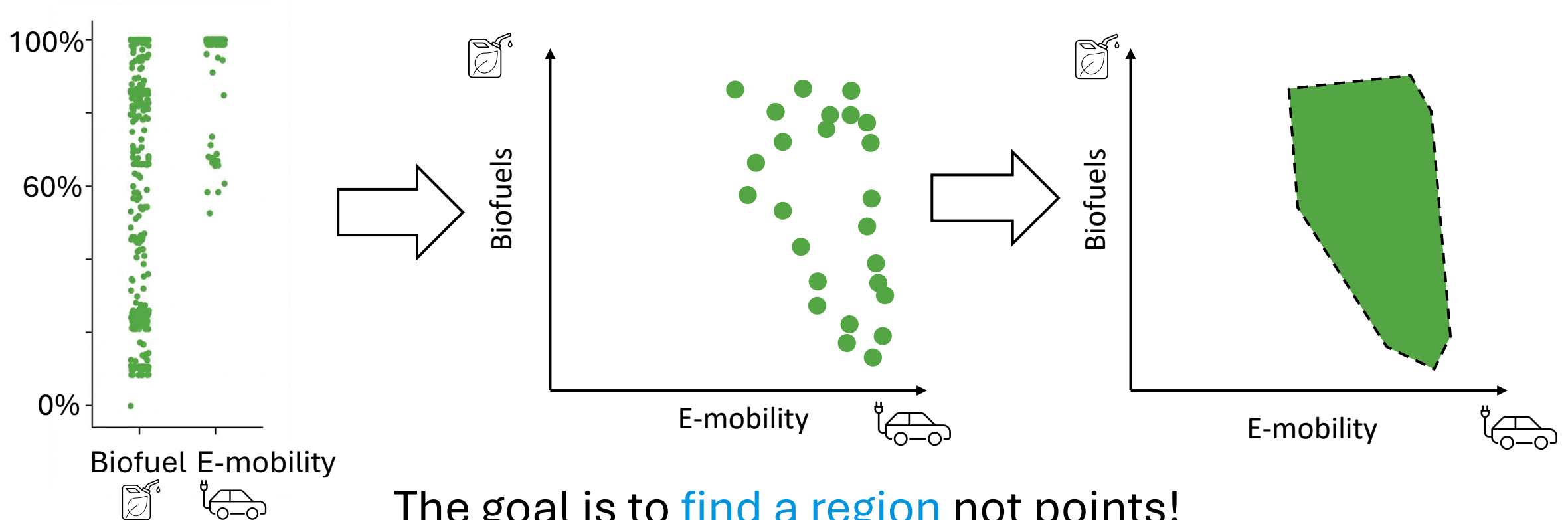
Biofuel ranges from fully used to avoided!



Pickering, B., Lombardi, F., & Pfenninger, S. (2022). Diversity of options to eliminate fossil fuels and reach carbon neutrality across the entire European energy system. *Joule*, 6(6), 1253-1276.
Berntsen, P. B., & Trutnevyte, E. (2017). Ensuring diversity of national energy scenarios: Bottom-up energy system model with Modeling to Generate Alternatives. *Energy*, 126, 886-898
Neumann, F., & Brown, T. (2021). The near-optimal feasible space of a renewable power system model. *Electric Power Systems Research*, 190, 106690.

Finding near-optimal solutions

What do we really want to do?



The model is a linear program → the region is a polytope!

How to find the near-optimal region?

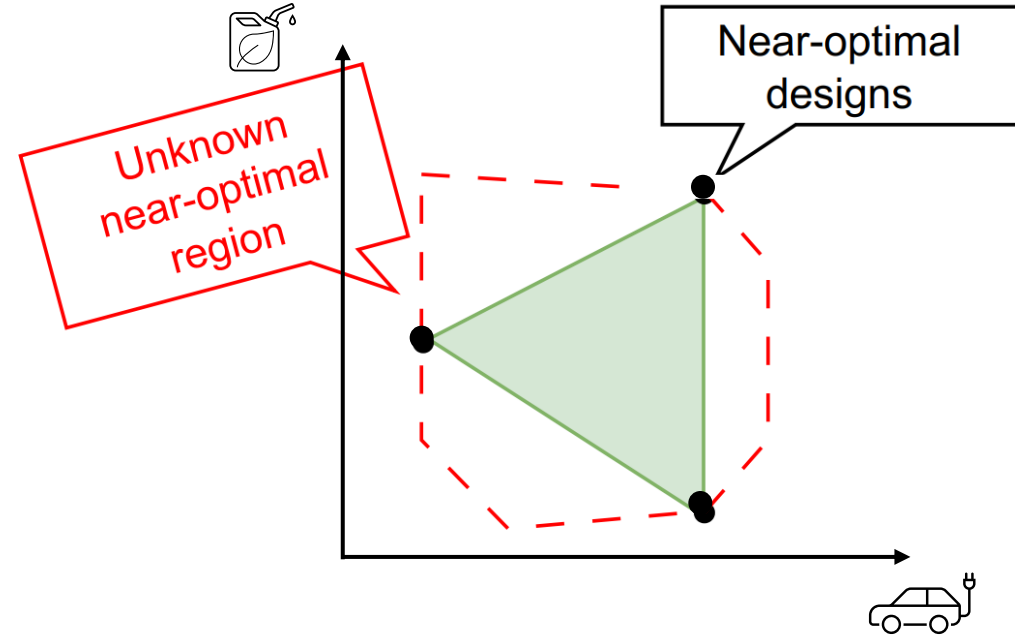
Exploiting convexity

We have near-optimal solutions

- Can we say anything about the near-optimal region?

The model is convex

- Everything in the **convex hull** is also near-optimal



We can define an inner approximation

What can we do with the information we have?

Exploiting convexity

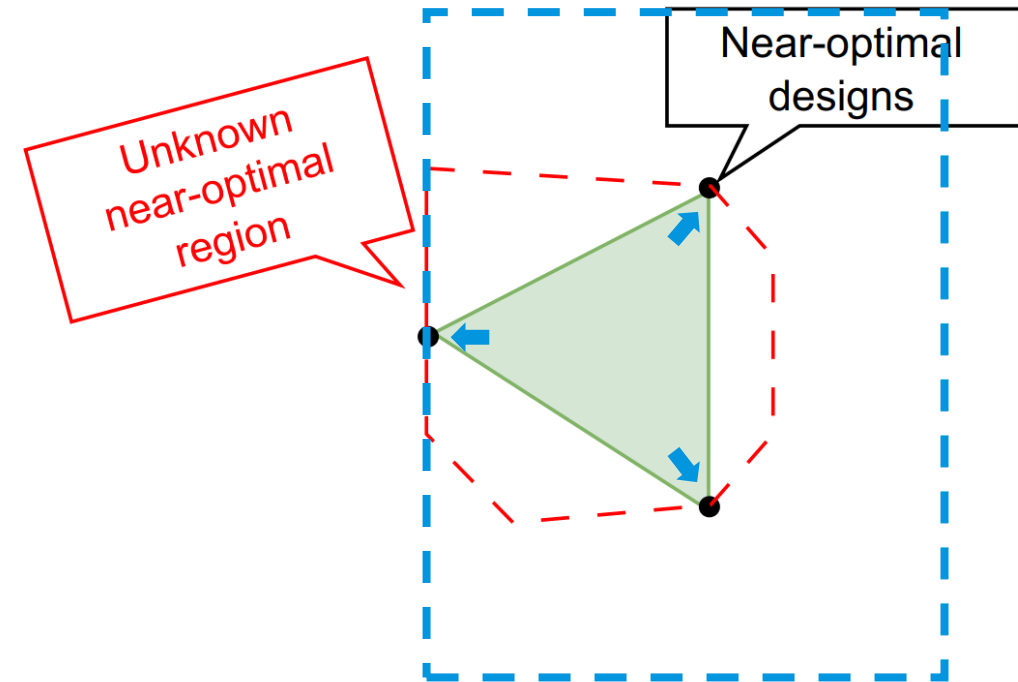
The model is convex

- The **convex hull** of points is also near-optimal

Can we do more?

We have **variable bounds**

We have **search directions**



What can we do with the information we have?

Exploiting convexity

The model is convex

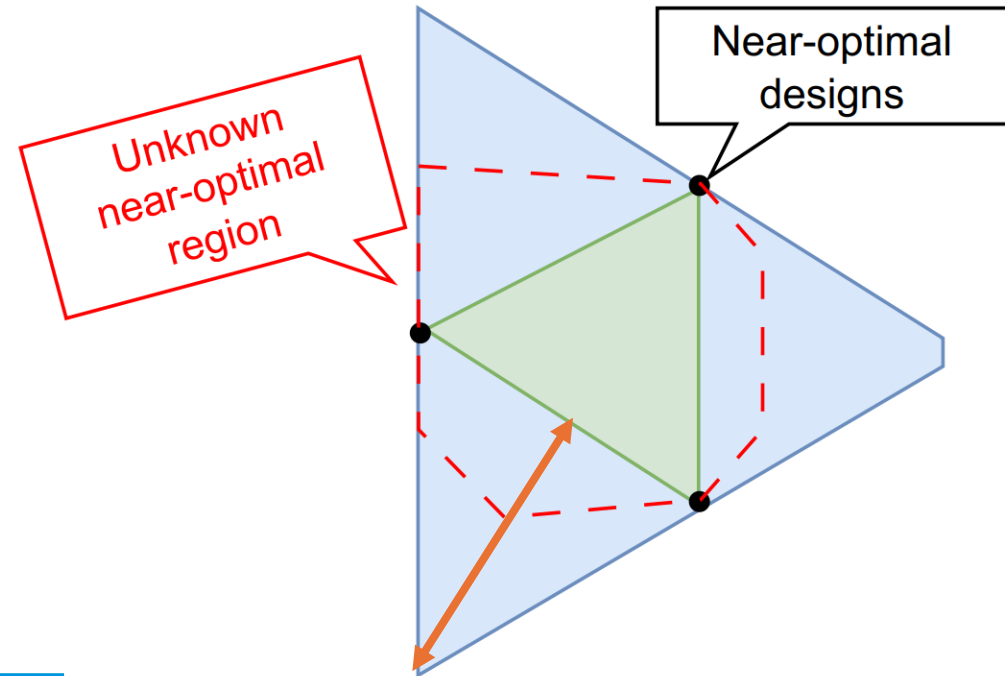
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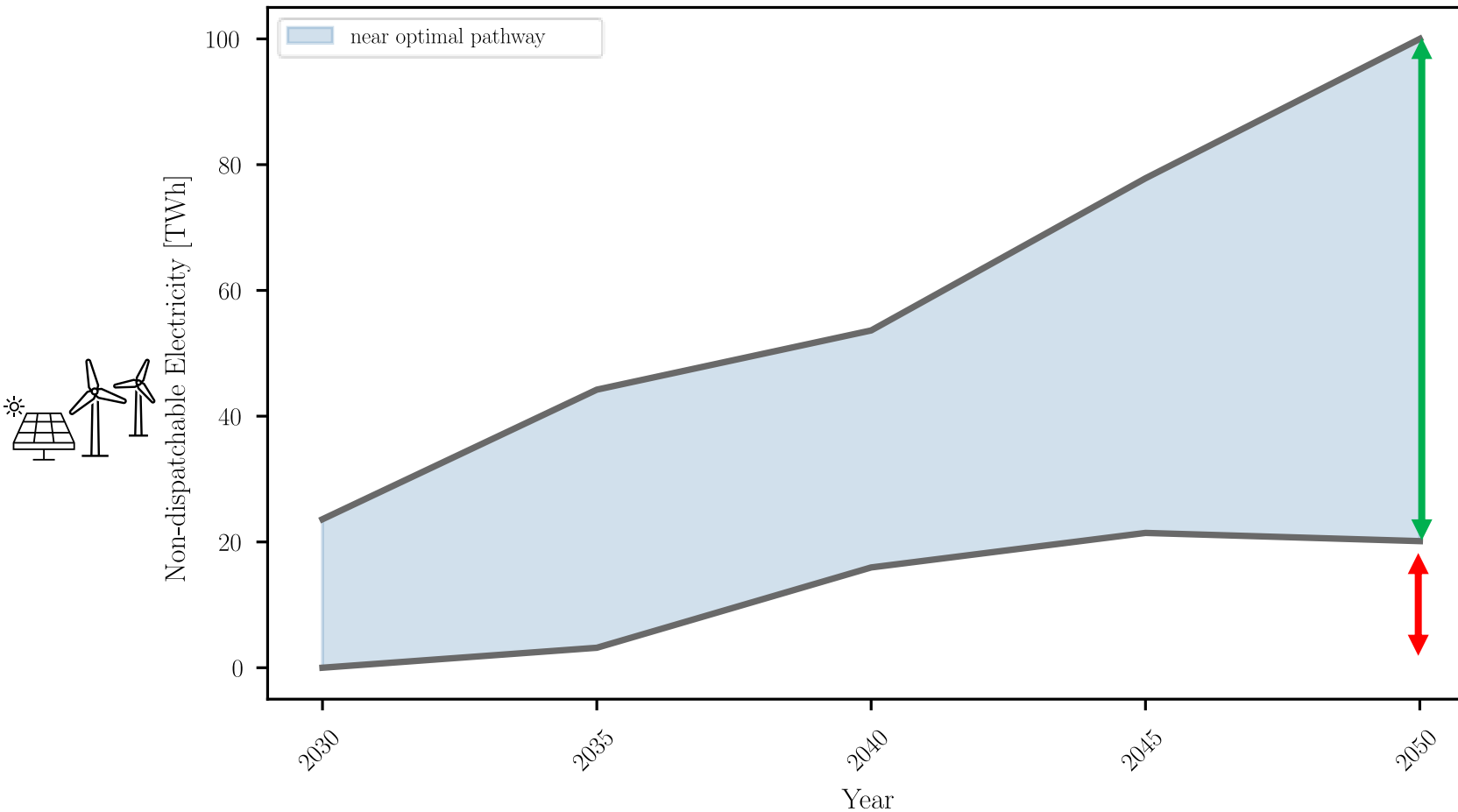
We can define **inner and outer approximations**



We can measure exploration

Near-optimal regions allow for efficient “what-ifs”

Study: transition of Switzerland within +2 billion €



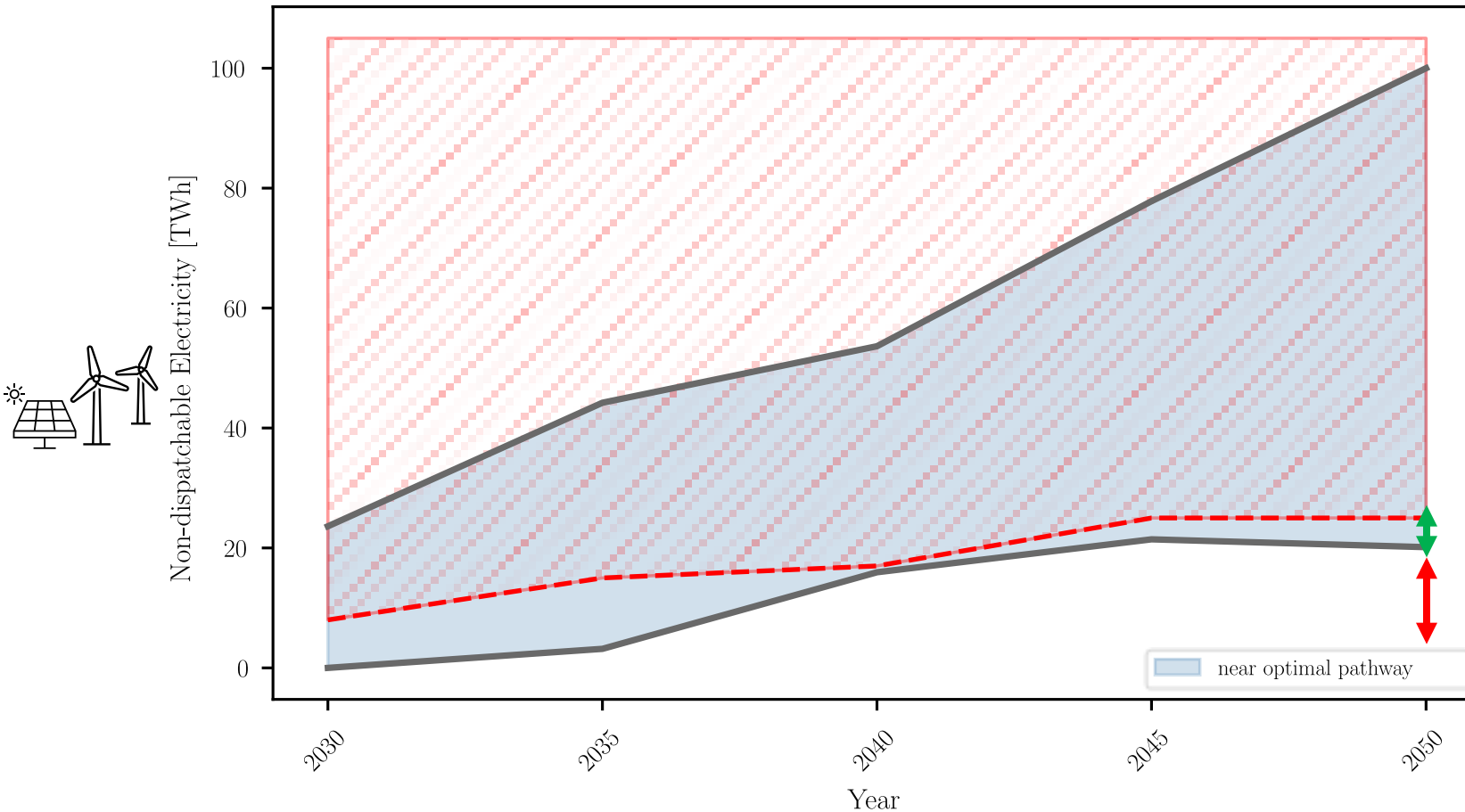
What if renewables are limited?

Large range in the deployment of renewables

Renewables *must* be deployed

Near-optimal regions allow for efficient “what-ifs”

Study: transition of Switzerland within +2 billion €



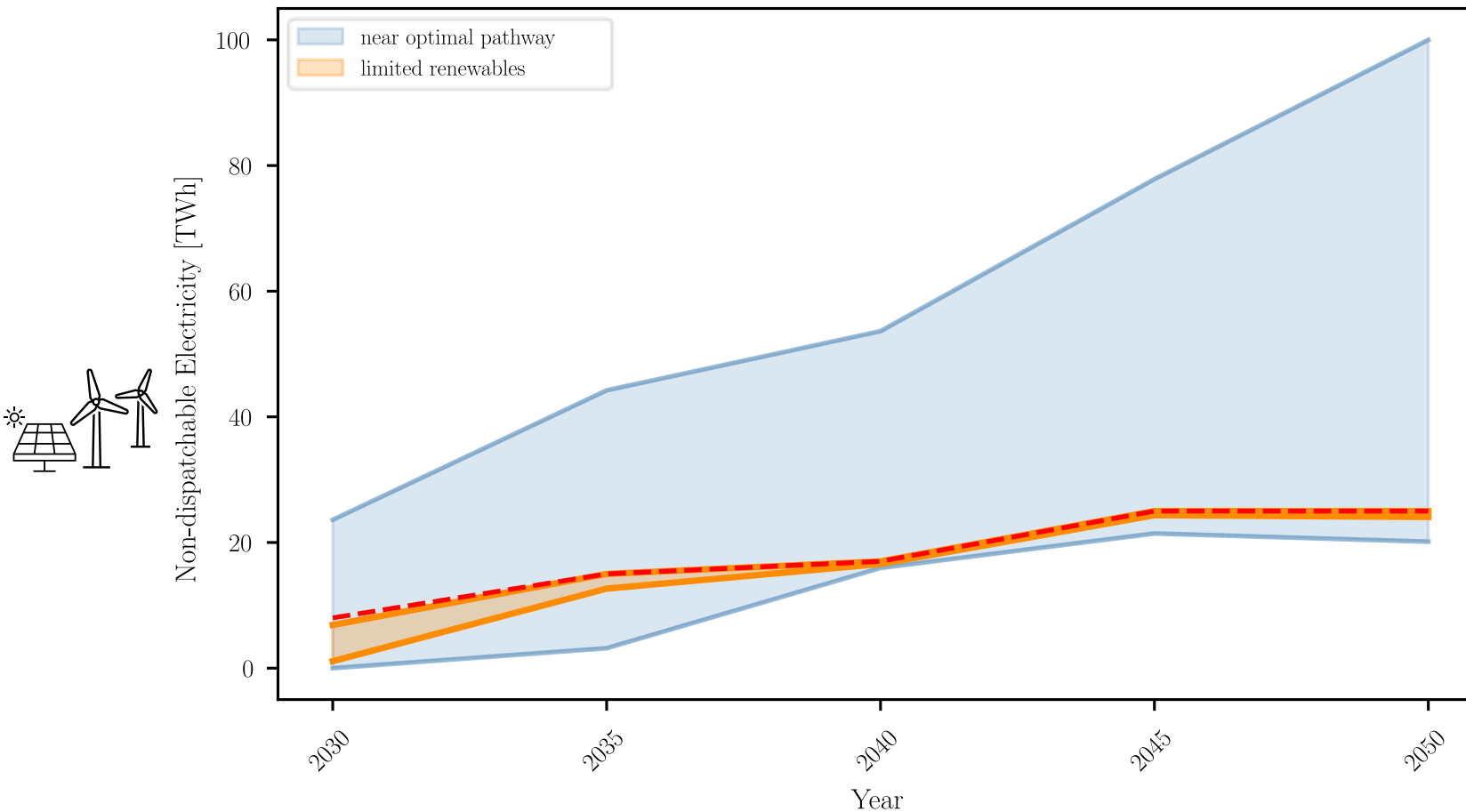
What if renewables are limited?

Reduced range in the deployment of renewables

Renewables *must* be deployed

Near-optimal regions allow for efficient “what-ifs”

Study: transition of Switzerland within +2 billion €

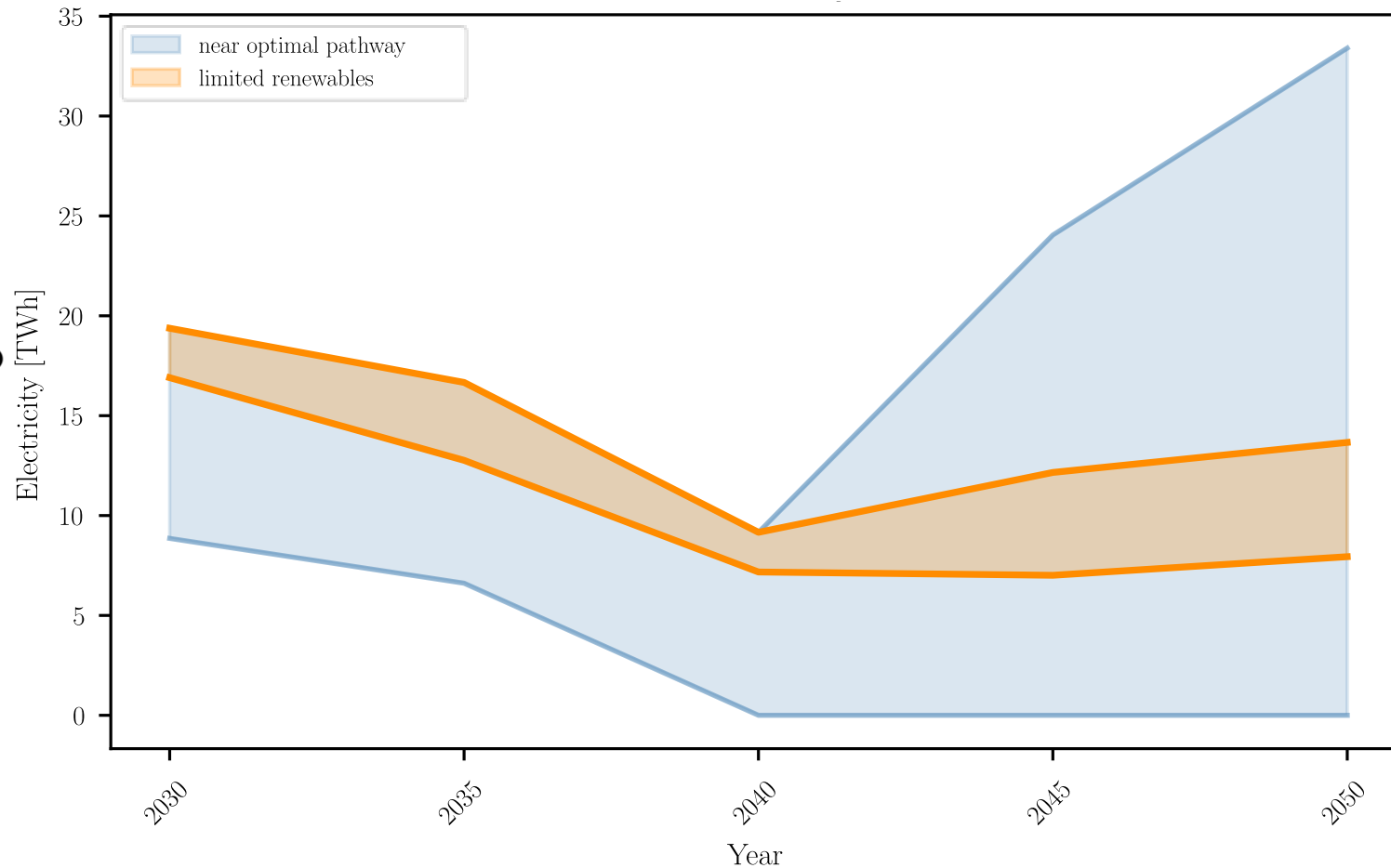


What if renewables are limited?

- The upper bound on renewables implies a lower bound!
- How are other technologies influenced?

Near-optimal regions allow for efficient “what-ifs”

Secondary effects



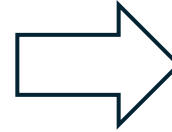
What if renewables are limited?

- The upper bound on renewables implies a lower bound on nuclear!

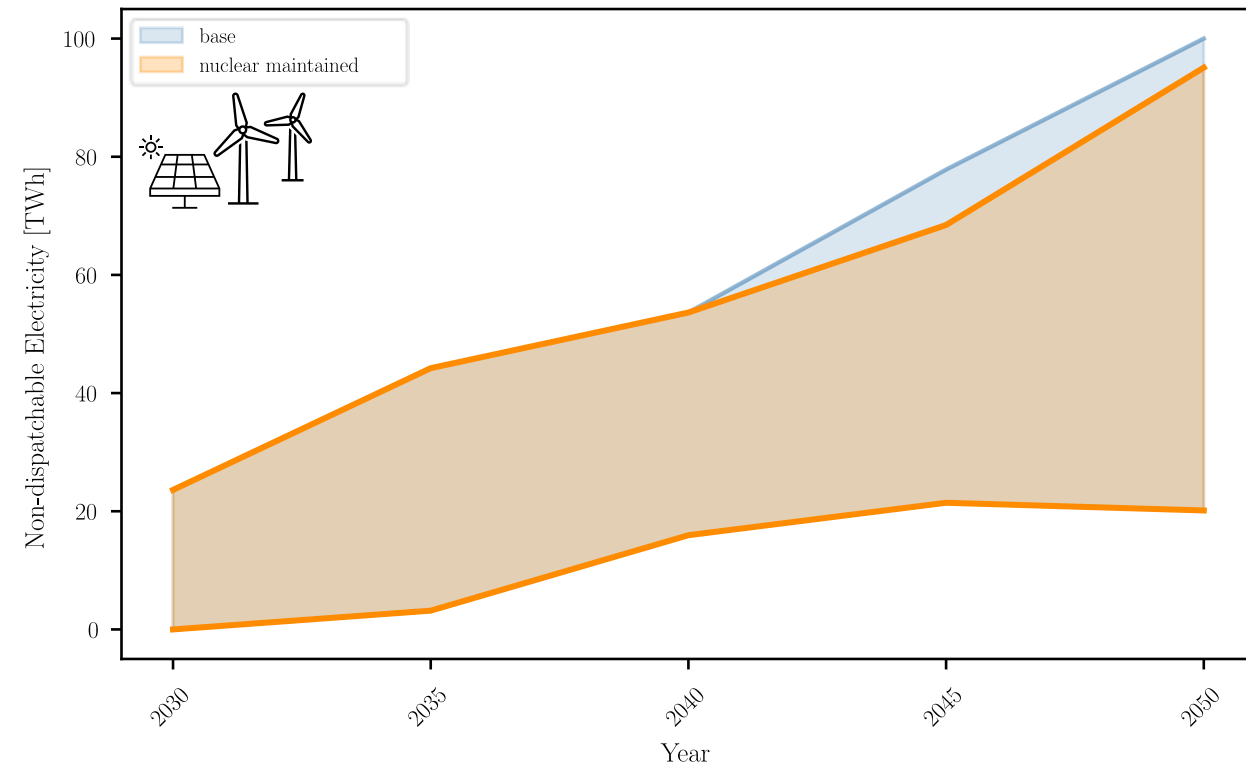
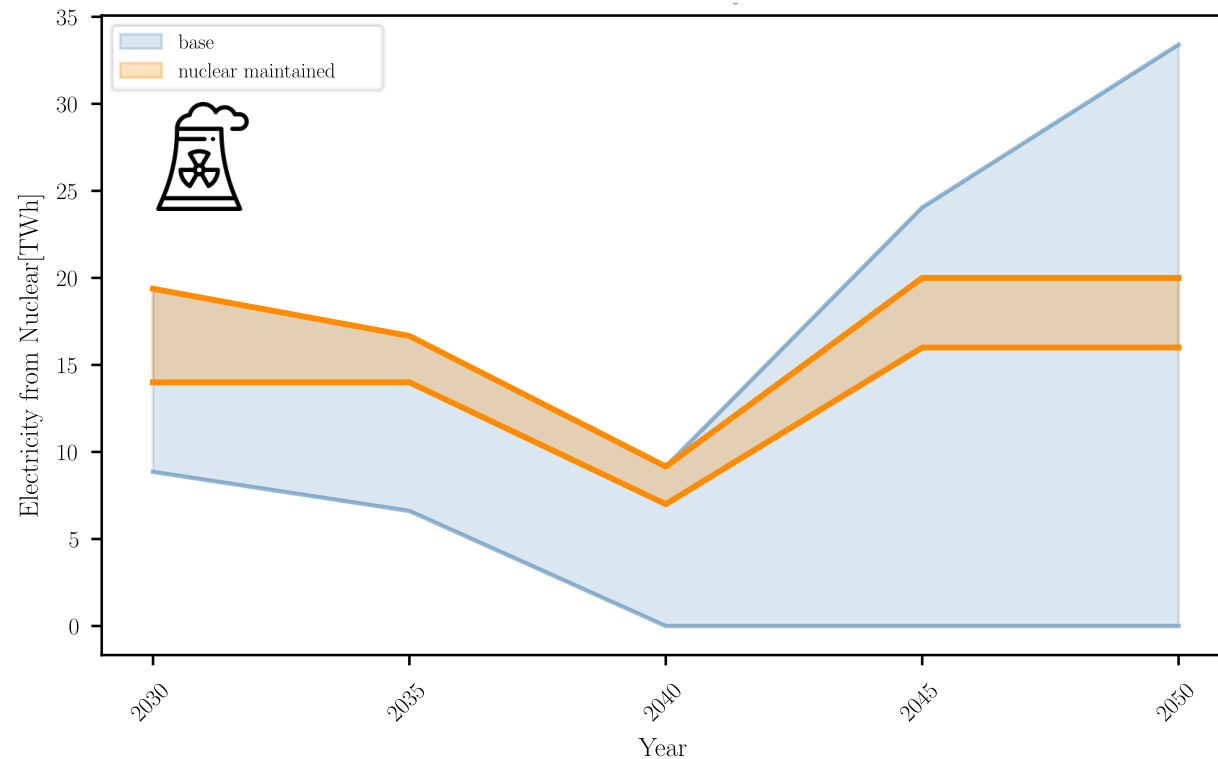
Near-optimal regions allow for efficient “what-ifs”

Restrictions are not symmetric

Nuclear production
maintained

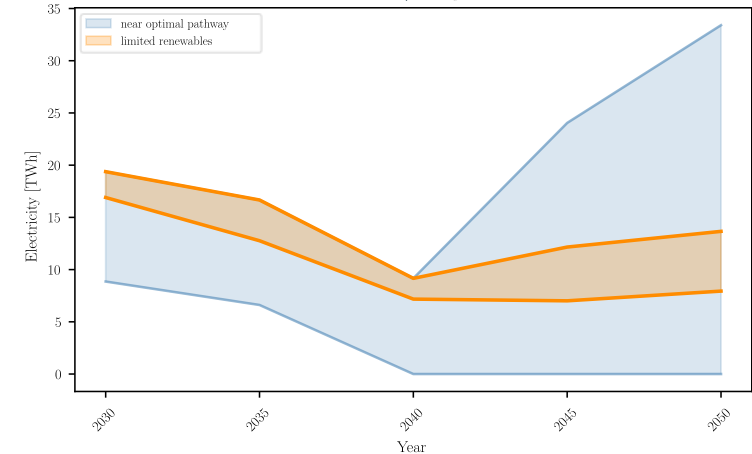
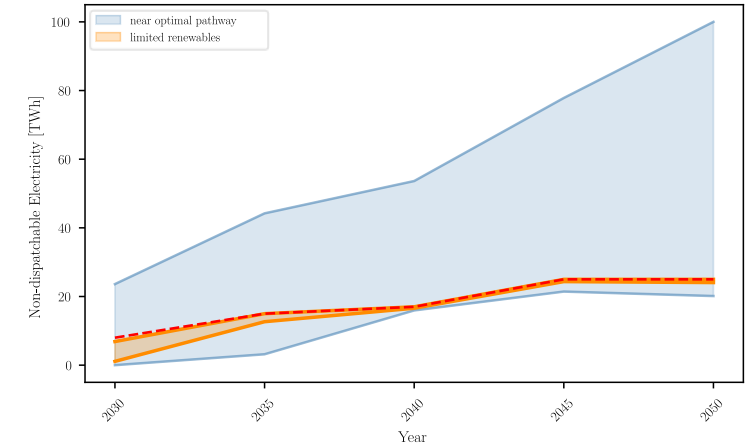


Implies a slight limit in renewable deployment



Conclusions

- Decision makers want more than the cost-optimal solution
- We efficiently find the near-optimal space through polytope approximation
- Once found we can perform what-ifs, scenario analysis, and **at negligible computational cost** (i.e. without running the system model)



Questions & comments?
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